Quantum Thermal Architecture: A Mode-Separated Validation Framework for Cryogenic LCVD and NV-Based Helium Isotope Sensing

[Author list]

Manuscript version v4.0 — June 6, 2026

GLOBAL STATUS: Current: BLOCKED (required hardware not installed; required measurements not performed). Post-install forecast: CONDITIONAL ($\tau_c \geq 292\,\mu$s AND $C_{\text{contr}} > 0$ at 10 mK required). Validated system: NOT AVAILABLE. Breakthrough claim: NOT MADE. Full-cycle MC (current): 0.0%. PASS count: 0.

Abstract

This work proposes a same-chamber, mode-separated cryogenic validation framework combining an LCVD-capable process mode (Mode B) with NV-center-based helium isotope sensing (Mode D) [1, 4]. **The framework is not experimentally validated.** The current state is BLOCKED because required hardware is not installed and required measurements are missing.

LCVD growth and NV sensing are mutually exclusive, hardware-interlocked operating modes (enforced by IL-01, IL-02, IL-14; see Section on interlocks). They do not operate simultaneously. The simulation is an assumption-tracked forecast model, not a first-principles proof. No gate reaches PASS without verified in-system measurement.

Detection feasibility is jointly gated by two co-equal scientific unknowns:

1. τ_c of adsorbed ^{3}He on the actual F-terminated diamond surface at 10 mK [11, 12] [UNKNOWN; canonical threshold $\tau_c \geq 292\,\mu$s].
2. C_{contr} at 10 mK under actual NV readout conditions [4, 9] [UNKNOWN].

If $C_{\text{contr}} = 0$ or too low at 10 mK, detection fails regardless of τ_c.

Unresolved engineering bottlenecks include: G_{eff} of the Ag-sinter/diamond interface [**ASSUMED**], residual gas suppression (bakeout not executed) [**NOT_INSTALLED**], cryotrap [**DESIGN_SPECIFIED**], NEG pump [**DESIGN_SPECIFIED**], optical absorption η_{abs} [**ASSUMED**], microwave dissipation P_{mw} [**ASSUMED**], vibration PSD at NV position S_{vib} [**MANUFACTURER_SPEC**], NV depth d_{NV} [**ASSUMED**], T_2^* [**ASSUMED**], surface spin density n_s [**ASSUMED**], and mode interlock behavior [**DESIGN_SPECIFIED**].

Gate counts: 0 Pass | 47 conditional | 0 Fail | 2 unknown | 23 blocked | 11 derived_check (D9 gas kinetics; 10 non-lumped multiphysics numerical checks).

Canonical gate-table summary (single source of truth, generated by `qta_full_sim.py`): 83 total gates — 0 PASS, 47 CONDITIONAL, 23 BLOCKED, 2 UNKNOWN, 11 DERIVED_CHECK. Canonical τ_c threshold: 292 μs (v3.3). The 27.728 μs value (v3.0) is SUPERSEDED, NOT_CANONICAL, NOT_LIVE_GATE_LOGIC.

1 Scope and Non-Claims

This manuscript does **not** claim:

- working millikelvin LCVD hardware
- demonstrated NV helium spin-noise detection
- mode-switched LCVD and NV sensing
- validated in-situ growth at 10 mK
- verified τ_c for ^{3}He on the actual diamond surface
- verified C_{contr} at 10 mK
- verified Ag-sinter G_{eff}
- verified residual gas suppression
- verified QCM coverage tracking
- verified optical absorption or transient recovery
- a breakthrough, Nobel-worthy result, or DARPA-ready system
- a validated engineering system
- a proof of feasibility

Correct category: blocked conditional validation framework.

The framework defines a mode-separated architecture and a program of experiments required to test whether the sensing premise survives contact with physical reality.

2 Canonical Mode Map

The architecture enforces strict temporal separation:

Mode	Label	Description
A	BASELINE/STABILIZATION	Cryogenic baseline at the operating se point; thermal/vacuum/NV-baseline verific tion; readiness for any exposure or processin Sensing OFF. LCVD OFF.
B	MATERIAL_PROCESSING/LCVD_GROWTH	LCVD on; CH_4 or other precursor preser fs-laser processing; pulsed molecular-beam d livery; sensing off; heat switch open. Heliu ABSENT.
C	ISOLATION_PURGE_THERMAL_RECOVERY	Growth inputs off; gas lines isolate cryopump/baffle residual species; sar ple thermally relaxes; vibration settle RGA/QCM/witness-coupon checks.
D	SENSING/MEASUREMENT	LCVD off; precursor below threshold; H 3 dosed; NV ODMR / Ramsey at the m likelvin sensing condition.

LCVD growth and NV sensing are not simultaneous. They occur as mutually exclusive, hardware-interlocked modes inside the same cryogenic chamber. Mode B and Mode D cannot run together. Mode C exists to isolate contamination and allow thermal recovery before Mode D begins; Mode A is the cryogenic baseline/stabilization state.

All mode transitions and interlocks are [**DESIGN_SPECIFIED**] unless physically implemented and tested.

3 Evidence Status Vocabulary

blocked Required hardware, measurement, or validation infrastructure missing; result cannot be interpreted.

conditional Model forecasts feasibility only if unverified assumptions later measure correctly.

unknown Parameter is too unknown to bound; no reliable forecast possible.

derived_check Internal mathematical or gas-kinetic check is self-consistent; not a physical system pass.

Fail Modeled or measured value violates threshold.

Pass Allowed *only* after verified in-system measurement under relevant conditions.

Absolute rule: No gate reaches PASS from [**ASSUMED**], UNKNOWN, [**LITERATURE_BOUND**], [**MANUFACTURER_SPEC**], [**DESIGN_SPECIFIED**], or [**NOT_INSTALLED**] inputs.

4 Committed Mechanical and Thermal Architecture

4.1 Rejected Baseline: Direct Vespel SP-22 Supports

Direct Vespel SP-22 polyimide supports are **rejected** and are not part of the committed architecture. They create an unacceptable direct conductive path from 4 K to the 10 mK sample node. All Vespel SP-22 calculations appear in Appendix B for reference only.

4.2 Committed Support Architecture

The committed sample-support architecture replaces direct Vespel rods with a hybrid Kevlar 49 / G-10CR intercepted suspension. Direct Vespel support is rejected and is not part of the current design.

Committed elements [**DESIGN_SPECIFIED**]:

- Kevlar 49 tension members ($\kappa_{\text{Kevlar}} \approx 5\times10^{-4}$ W/m/K at 10 mK [3]) [**DESIGN_SPECIFIED**]
- G-10CR fiberglass flexures for lateral constraint [**DESIGN_SPECIFIED**]
- Staged OFHC copper thermal intercepts at 4 K, 1 K, 100 mK, 10 mK [**DESIGN_SPECIFIED**]
- Gold-plated copper anchor blocks [**DESIGN_SPECIFIED**]
- Ag-sinter thermal interface to sample stage [**DESIGN_SPECIFIED**] (45 cm^2 [**ASSUMED**])

Estimated support heat load: $P_{\text{support}} \approx 0.167$ nW (Kevlar/G-10CR intercepted; [**ASSUMED**] — step-response not performed).

Gate status: CONDITIONAL (E05, E06; hardware not fabricated).

5 Mode-Local Secondary Thermal Feedback (Mode D Only)

Within Mode D sensing, the sample temperature obeys the iterative relation:

$$T_s = T_{\text{fridge}} + \frac{P_{\text{total}}(T_s)}{G_{\text{eff}}} \tag{1}$$

where G_{eff} [**ASSUMED**] $= 10^{-5}$ W/K (Kapitza model [2, 1]; Ag sinter not fabricated) and P_{total} includes all Mode D thermal loads: conduction through wiring (P_{cond} [**ASSUMED**]), optical average load (P_{opt} [**ASSUMED**]), microwave dissipation (P_{mw} [**ASSUMED**]), support conduction (P_{support} [**ASSUMED**]), vibration (P_{vib} [**MANUFACTURER_SPEC**]), and radiation (P_{rad}).

At design parameters: $T_{\text{fridge}} = 10$ mK [**MANUFACTURER_SPEC**], $P_{\text{total}} = 4131$ pW [**ASSUMED**], $T_s = 10.413$ mK [**ASSUMED**]. Margin $\approx 4 \times 10^4$ [**ASSUMED**] — requires G_{eff} verification.

Individual Mode D load terms may exhibit mode-local secondary feedback (a higher T_s increases P_{rad} and P_{He}, closing a small perturbative loop). This is evaluated within

Mode D only. **No cross-mode thermal coupling between Mode B LCVD loads and Mode D loads occurs; LCVD and sensing are temporally separated.**

All thermal budget gates (D3, D4, D5): CONDITIONAL — G_{eff} and all load terms are [**ASSUMED**].

6 Vacuum, Residual Gas, and Suppression Strategy

Residual gas suppression is a required implementation condition, not a demonstrated result.

Current state: bakeout not executed; NEG not installed; cryotrap not installed. Gates D10a, D10b, E01, E04: BLOCKED.

Required implementation (all [**DESIGN_SPECIFIED**]):

1. 250°C / 48 h UHV bakeout of chamber and lines [16, 17]
2. SAES St707 NEG pump activation post-bakeout [19]
3. 77 K charcoal cryotrap [18]
4. RGA all-species verification (FC-corrected for sensing-surface pressure) [20]

Pre-bakeout $P_{\mathrm{H_2}} \approx 10^{-10}$ Pa $\Rightarrow \theta_{\mathrm{H_2}} \approx 0.32\%$ (above 0.1% threshold; BLOCKED until bakeout executed). Post-bakeout forecast: $\theta_{\mathrm{H_2}} \approx 0.003\%$ (CONDITIONAL; [**ASSUMED**] — not measured).

7 Independent Gas Line Architecture

Required isolation architecture [**DESIGN_SPECIFIED**]:

- Independent ^{3}He sensing line (dedicated; no dead volume shared with CH_4)
- Independent ^{4}He reference line (dedicated control)
- CH_4 / LCVD precursor line (Mode B only; isolated during Mode D)
- Pumpout / purge line (Mode C operation)
- Line-specific shutoff valves, pressure gauges, and cryogenic-compatible hardware
- No backstreaming path from CH_4 to helium sensing lines

Interlock IL-02: CH_4 and He-3 valves may not be open concurrently. All gas lines: [**DESIGN_SPECIFIED**]. Gate status: CONDITIONAL (E07, A4).

PMB / shuttered cryo-baffle for helium dosing: [**DESIGN_SPECIFIED**]; not installed. Provides temporal control and isotope-contrast separation. Status of PMB: CONDITIONAL (E08).

8 QCM Coverage Proxy

A 5 MHz AT-cut QCM adjacent to the diamond surface is proposed as a coverage proxy. Sauerbrey sensitivity [21]: $C_f = 2f_0^2/(\rho_q v_q) = 5.66 \times 10^5$ Hz/(g/m^2) $= 0.057$ Hz/(ng/cm^2). He-3 monolayer (1.65 ng/cm^2): $\Delta f \approx 0.09$ Hz. CH_4 monolayer (estimated): $\Delta f \approx 0.15$ Hz.

QCM-based coverage inference remains conditional until its frequency stability and dose correlation are demonstrated under the actual cryogenic operating conditions.

Requirements: sub-0.1 Hz stability; vibration isolation; thermal drift separation; cryogenic calibration. Status: **[DESIGN_SPECIFIED]**; not installed. Gate B5: CONDITIONAL.

9 NV Helium Isotope Detection Model

NV centers in Mode D respond to the stochastic magnetic field produced by diffusing ^{3}He nuclear spins [7, 6]. The decoherence rate contribution is [4]:

$$\Gamma_{\mathrm{DC}} = \left(\frac{\mu_0}{4\pi}\right)^2 \frac{\gamma_{\mathrm{NV}}^2\, \gamma_{^3\mathrm{He}}^2\, \hbar^2\, n_s\, \tau_c}{d^6} \propto \frac{S(\omega{=}0,\ \tau_c)}{d^6} \tag{2}$$

where $S(\omega, \tau_c) = 2\Gamma_0\tau_c/(1+\omega^2\tau_c^2)$ is the Lorentzian spectral density of the surface magnetic noise, n_s **[ASSUMED]** is the areal spin density, d **[ASSUMED]** is the NV–surface separation, and τ_c UNKNOWN is the ^{3}He surface correlation time.

Measurement protocol uses ^{4}He as a non-magnetic reference:

$$\Delta\Gamma = \Gamma_{^3\mathrm{He}} - \Gamma_{^4\mathrm{He}} \tag{3}$$

Detection requires $\mathrm{SNR} = \Delta\Gamma/\delta\Gamma \geq 5$ [10], where $\delta\Gamma \propto (C_{\mathrm{eff}}\, T^*_{2,\mathrm{eff}}\, \sqrt{N})^{-1}$.

9.1 τ_c and C_{contr}: Co-Equal Scientific Bottlenecks

Detection feasibility is jointly gated by τ_c and C_{contr} at 10 mK.

τ_c **[unknown]** The correlation time of ^{3}He adsorbed on F-terminated diamond at 10 mK is completely unknown. Graphite-surface analogs [11, 12, 13] provide plausibility only; surface termination fundamentally affects dynamics. Canonical threshold (v3.3; combined SNR≥ 5 + $\varepsilon_{\mathrm{thermo}} < 1\%$ including pulse dephasing):

$$\tau_c \geq 292\,\mu\mathrm{s} \tag{4}$$

Older threshold 27.7 μs (v3.0; no pulse dephasing): **[SUPERSEDED v3.0]**. See Appendix A.

C_{contr} **at 10 mK [unknown]** NV ODMR contrast at 10 mK under pulsed 532 nm excitation is unknown [9, 8]. Room-temperature values (typically 0.05–0.15) may not hold at 10 mK due to charge-state instability under cryogenic laser irradiation [5].

If $C_{\mathrm{contr}} = 0$ or too low at 10 mK, detection fails regardless of τ_c.

Gate D12 [UNKNOWN]: C_{contr} at 10 mK not measurable without ODMR at operating temperature. Gate D13 [CONDITIONAL]: detection SNR requires $\tau_c \geq 292\,\mu$s **and** $C_{\mathrm{contr}} > 0$.

10 NV Sample Requirements

All NV parameters are [**ASSUMED**] or UNKNOWN unless measured on the actual sample:

Parameter	Value	Status	Required measurement
d_{NV}	10 nm	[**ASSUMED**]	SIMS or PL depth
T_2^*	10 μs	[**ASSUMED**]	Ramsey in cryostat
C_{contr} at 10 mK	UNKNOWN	UNKNOWN	ODMR at 10 mK
η_{col}	0.0635	[**ASSUMED**]	In-cryostat calibration
n_s (surface spins)	$3.3 \times 10^{18}\,\mathrm{m}^{-2}$	[**ASSUMED**]	TPD or QCM on F-diamond
Surface termination	F-terminated	[**DESIGN_SPECIFIED**]	XPS + AFM post-termination

11 Optical and Femtosecond Laser Constraints (Mode B and Mode D)

Two distinct optical uses: (1) Mode B femtosecond laser for LCVD (proposed; [**DESIGN_SPECIFIED**]); (2) Mode D CW or pulsed laser for NV readout (proposed; [**DESIGN_SPECIFIED**]). These are temporally separated.

LCVD femtosecond laser parameters [**ASSUMED**]: $E_{\mathrm{pulse}} = 10\,\mathrm{fJ}$, $f_{\mathrm{rep}} = 1\,\mathrm{MHz}$, $\tau_{\mathrm{pulse}} = 100\,\mathrm{fs}$, $r = 361\,\mathrm{nm}$, $\eta_{\mathrm{abs}} = 0.05$ [**ASSUMED**]. Fluence: $F \approx 2.4 \times 10^{-2}\,\mathrm{J/m^2} \ll$ damage threshold $\sim 10^4\,\mathrm{J/m^2}$ [22, 23] [**LITERATURE_BOUND**].

Low average power does not by itself validate femtosecond-pulse safety; transient fluence, nonlinear response, absorption, and recovery must be measured.

Pulse gating: AOM switching time ($\sim$100 ns–1 μs) cannot gate 100 fs pulse envelopes. If individual pulse gating is required, a Pockels cell or equivalent ultrafast shutter is needed. Status: [**DESIGN_SPECIFIED**]. Thermal recovery: $\tau_{\mathrm{thermal}} \approx 42\,\mathrm{ns}$ (Debye model; [**ASSUMED**]) $\ll$ 1 ms rep period.

Gate D7 [CONDITIONAL]: transient heating conditional on η_{abs} [**ASSUMED**].

12 Microwave Delivery and Heating

Mode D only. CPW geometry [**DESIGN_SPECIFIED**]; P_{mw} at sample $\approx 1\,\mathrm{nW}$ [**ASSUMED**] (room-temperature source + estimated cryogenic attenuation; not measured at base). Heating estimate: $\Delta T_{\mathrm{mw}} \approx 100\,\mu\mathrm{K}$ [**ASSUMED**]. Threshold: $< 104\,\mu\mathrm{K}$ (1% of T_s).

Microwave and vibration loads remain conditional until measured at the sample/NV position under Mode D operating conditions.

Gate D16 [CONDITIONAL]: ΔT_{mw} marginal; both P_{mw} and G_{eff} are [**ASSUMED**].

13 Vibration Constraints

S_{vib} at NV position: $10^{-10}\,\text{m}^2/\text{Hz}$ [**MANUFACTURER_SPEC**] (Oxford Triton specification; not measured at NV location in this setup). Vibration-to-heat conversion requires mode-local model within Mode D.

Gate D17 [CONDITIONAL]: Helmholtz geometry [**DESIGN_SPECIFIED**]; S_{vib} not measured in situ.

14 Helium Molecular Flow (Mode D)

^{3}He transport in the dosing geometry is in the molecular-flow (Knudsen) regime. At dosing conditions ($T = 4\,\text{K}$, $P = 1\,\mu\text{Pa}$): $\lambda_{\text{He}} \approx 0.46\,\text{m} \gg L_{\text{char}} = 10\,\text{mm}$, giving $\text{Kn} \approx 46$ [14, 15].

Gate D9 status: DERIVED_CHECK — gas-kinetic check is self-consistent; confirmed by formula from standard references; **not** a physical system PASS (would require actual He-3 flow measurement in the built system).

15 Mode B Cryogenic LCVD / Surface-Processing Feasibility Limits

Mode B is a proposed pulsed surface-processing mode, not a demonstrated capability. It is mutually exclusive with Mode D sensing by hardware interlock (IL-01, IL-02, IL-14). Helium is forbidden in Mode B (Gate A14, IL-14).

Gas-phase truth

Normal LCVD requires a gaseous or beam-delivered precursor at the reaction site. At millikelvin substrate and shield temperatures, methane and other hydrocarbon precursors will *cryocondense* on cold surfaces unless delivered through a controlled warm/differentially pumped molecular beam path. Therefore this work **does not claim conventional chamber-fill LCVD at 10 mK**. Mode B is a proposed pulsed, local, beam-fed surface-processing mode whose deposition yield per pulse must be measured.

Mode B laser configuration (separate from Mode D NV readout)

Mode B LCVD laser parameters are *not* the same as Mode D NV readout parameters. The package configuration vectors are kept separate:

- **Mode B LCVD optics**: E^A_{pulse} [**DESIGN_SPECIFIED**], f^A_{rep} [**DESIGN_SPECIFIED**], τ^A_{pulse} [**DESIGN_SPECIFIED**], spot radius [**DESIGN_SPECIFIED**], η^A_{abs} [**DESIGN_SPECIFIED**], precursor species [**DESIGN_SPECIFIED**], deposition yield per pulse UNKNOWN, local surface temperature during deposition UNKNOWN.
- **Mode D NV readout**: E^D_{pulse} [**DESIGN_SPECIFIED**], f^D_{rep} [**DESIGN_SPECIFIED**], η^D_{abs} [**ASSUMED**], T^D_{peak} [**ASSUMED**], τ^D_{recovery} [**ASSUMED**].

Mode D gates D6/D7 use only the Mode D vector. Mode B gates A6–A13 use only the Mode B vector. The two vectors are never mixed in a single gate.

Mode B LCVD feasibility gates

- A6 Precursor remains beam-delivered/gaseous until reaching the target; does not act as a chamber-filling gas. CONDITIONAL until measured.
- A7 Precursor cryocondensation on cold shields and on the sensing zone is below limit. CONDITIONAL until measured by witness coupon and RGA.
- A8 Local surface temperature during the deposition pulse is *measured*, not assumed; photolysis or pyrolysis threshold is verified at that surface temperature. BLOCKED until pump–probe or thermometric measurement is available.
- A9 Deposition yield per pulse is *measured* by witness coupon / QCM / AFM / Raman / XPS. BLOCKED until measured.
- A10 Growth-zone contamination does not reach sensing-zone coupon. CONDITIONAL until measured by witness coupon and RGA after each Mode B pulse train.
- A11 Byproducts and unreacted precursor pump below RGA thresholds *before* Mode D begins. BLOCKED until RGA is installed and all-species threshold is verified.
- A12 Mode B heat is removed by a dedicated non-MC thermal dump path (Section 15.1). CONDITIONAL until the dedicated dump switch and dump-stage temperature sensors are installed and the heat balance is measured.
- A13 Repeated mode cycling (Mode B processing → Mode C recovery → Mode D sensing) passes fatigue and stress inspection (Section 15.2). CONDITIONAL until $N = 100$ qualification cycles are completed.
- A14 He-3 / He-4 film absent before Mode B. BLOCKED until verified in-situ by RGA, QCM, and pressure logging. Hardware interlock IL-14: LCVD_ON $\wedge$ HELIUM_FILM_PRESENT = IMPOSSIBLE. Rationale: adsorbed or superfluid helium can wet surfaces, block precursor access, alter thermal transport, alter NV optical/magnetic environment, and corrupt surface chemistry. Helium belongs only to Mode D sensing.

None of A6–A14 may reach PASS from assumptions, simulation convergence, or literature; each requires measured in-system data.

15.1 Mode B non-MC thermal dump architecture

The mixing chamber (MC) cannot remove Mode B laser heat in real time. The proposed Mode B thermal dump architecture is:

- **MC-protection switch**: superconducting heat switch held *open* during Mode B so the MC is decoupled from the growth zone.
- **Growth thermal dump switch**: an independent heat switch that *closes* during Mode B and connects the growth-zone carrier to a 1 K or 4 K dump stage (separate from the MC).
- **Radiation shutter stack**: closed during Mode B to protect the 10 mK region from optical and IR radiation from the growth zone.
- **Temperature sensors**: independent calibrated thermometers on the growth-zone surface, sample bulk, MC, 1 K stage, and 4 K stage, logged continuously during Mode B.

The minimal Mode B thermal solve is $T_{\text{growth\_surface}}(t)$, $T_{\text{sample\_bulk}}(t)$, $T_{\text{MC}}(t)$ driven by $P^A_{\text{laser}}(t)$ and $P^A_{\text{precursor}}(t)$ through $G_{\text{growth\_to\_dump}}$ with parasitic $G_{\text{growth\_to\_MC\_leak}}$. The requirement is that the MC heat leak stays below the MC cooling budget ($\ll 200\,\mu$W) while the growth-zone heat is absorbed by the non-MC dump. This package does not assert that the 10 mK MC removes LCVD heat in real time. All conductances $G_\cdot$ are [**DESIGN_SPECIFIED**] and must be measured before Mode B is run.

15.2 Thermal cycling, stress, and fatigue

Repeated A/B/C/D cycling stresses material interfaces. The package requires measurement (or, where unavailable, CONDITIONAL status) for: diamond / Ag-sinter CTE mismatch; Ag sinter cracking or delamination; Kevlar end-termination slip after repeated cycles; G-10CR flexure fatigue; copper anchor stress; wire and CPW fatigue at mode-switching cadence; shutter actuator vibration and wear; gas-line and feedthrough cycling.

Acceptance criteria: no cracks; no delamination; no conductance drift greater than threshold; no RGA contamination increase; no measurable alignment drift, after $N = 100$ cycles (first engineering qualification) and $N = 1000$ cycles (extended qualification). Status remains CONDITIONAL until the package is physically cycled.

Required Mode B validation evidence

Witness coupon + QCM + AFM + Raman + XPS + RGA records must be archived for *every* Mode B run. Mode D is permitted to begin only after the witness coupon, RGA, and thermometric records demonstrate that A11 (byproduct removal), A10 (cross-contamination), A14 (helium-film absence), and A12 (thermal recovery) are satisfied.

Final Mode B status: BLOCKED (hardware not installed) / CONDITIONAL post-installation. LCVD at 10 mK is not validated.

15.3 Speculative material extensions (future work only)

The following are **not** part of the current gate verdict and require independent experimental validation before being incorporated: isotopically pure C-13 diamond growth, graphene-on-diamond, CNT forests, graphene–diamond–CNT hybrids, or any post-silicon device concept.

16 Simulation Architecture

`qta_full_sim.py` implements a single-mode state vector (`ModeStateVector`) that computes, in order: (1) surface coverage estimates (θ_{H_2}, helium dose); (2) molecular transport (λ_{He}, Kn); (3) iterative thermal solve ($T_s = T_f + P(T_s)/G_{\text{eff}}$, converged to $< 1\,\mu$K); (4) optical transient (T_{peak}, F, η_{abs} [**ASSUMED**]); (5) detection model (Γ_{DC}, SNR, τ_c threshold); and (6) mode-local secondary load terms.

Simulation outputs are conditional forecasts. Numerical convergence does not imply physical validation. No gate reaches Pass from assumed inputs.

Run: `python3 qta_full_sim.py` Outputs written to `outputs/` directory.

16.1 Non-lumped 1D/2D reduced-order multiphysics layer

In addition to the lumped estimates above, QTA now includes a serious 1D/2D *non-lumped reduced-order multiphysics forecast layer* (`qta_multiphysics/`). It provides spatially-resolved forecast backends for thermal transport, moving-boundary laser absorption, gas/contamination transport, surface coverage, optical deposition, microwave heating, radiation leakage, vibration transfer, and the coupled Mode B → Mode C → Mode D recovery cycle. The 1D moving-boundary heat equation $\rho C_p(T)\,\partial_t T = \partial_z[k(T)\partial_z T] + Q_{\mathrm{laser}} + Q_{\mathrm{mw}} + Q_{\mathrm{bg}}$, with a Kapitza-type backside sink $-k\,\partial_z T|_L = \alpha_K(T_L^4 - T_{\mathrm{fridge}}^4)$, is solved by finite-volume method-of-lines with a stiff (BDF) integrator; the 2D axisymmetric backend solves the same physics in (r, z) with cylindrical $2\pi r$ weighting, a symmetry axis at $r = 0$, and a Gaussian radial beam profile.

1D is the canonical, fast, default non-lumped path; 2D axisymmetric is the serious spatial refinement; 3D is future work and is neither implemented nor claimed in this work. The layer is numerically self-checked (mesh convergence, source-energy conservation, Kapitza-sign, axis non-singularity, and a verified 2D→1D reduction to ∼1.4%); these are numerical self-consistency checks only. Every output of the layer is model-only and carries `measured_in_this_system = false`; the legacy lumped model is retained solely as a comparator. The layer contributes 20 additional gates, all CONDITIONAL/BLOCKED/DERIVED_CHECK — **none reaches Pass**. As with the lumped simulation, numerical convergence does not imply physical validation, and all multiphysics outputs remain forecast-only until hardware validation.

17 Gate Table and Current Verdict

Gate status is defined by the strictest supporting evidence. Full table in `results_gate_table.csv`. Selected critical gates:

Gate	Status	Canonical value	Blocking reason
D9	DERIVED_CHECK	Kn = 46	Gas kinetics only; not in-system verified
D10a	BLOCKED	Bakeout: not executed	E01: bakeout NOT executed
D10b	BLOCKED	$\theta_{\mathrm{H}_2} = 0.32\%$	Follows D10a block
D12	UNKNOWN	C_{contr}: UNKNOWN	ODMR at 10 mK not performed
D13	CONDITIONAL	τ_c: UNKNOWN	Requires $\tau_c \geq 292\,\mu$s AND $C_{\mathrm{contr}} > 0$
D5	CONDITIONAL	G_{eff}: ASSUMED	Ag sinter not fabricated
E01	BLOCKED	Bakeout: NOT INSTALLED	Must execute before D10a
E04	BLOCKED	RGA: NOT PERFORMED	Must verify residual gas
E11–E14	BLOCKED	Various calibrations	Not performed
A1–A5	CONDITIONAL	Various	Hardware **[DESIGN_SPECIFIED]**

Summary: 0 Pass | 39 CONDITIONAL | 0 Fail | 2 UNKNOWN | 21 BLOCKED | 1 DERIVED_CHECK

CURRENT SYSTEM: BLOCKED. Mode B prerequisites not executed. Full-cycle MC = 0.0% (logical consequence of 23 blocked gates, not a statistical result).
POST-INSTALLATION FORECAST: CONDITIONAL. Mode D forecast MC $\approx$ 33.8% (HYPOTHETICAL; post-installation; τ_c log-uniform $[1\,\mathrm{ns}, 100\,\mathrm{ms}]$). Dominant failure: $\tau_c < 292\,\mu$s (66.2% of failed samples). CO-EQUAL BOTTLENECK: C_{contr} at 10 mK unknown.
VALIDATED SYSTEM: NOT AVAILABLE.

18 First Decisive Experiment

The first decisive experiment is not a full QTA demonstration.

Question: Can shallow NV centers at 10 mK retain usable contrast and distinguish ^{3}He from ^{4}He surface-induced decoherence under controlled exposure?

Required measurements:

1. C_{contr} at 10 mK via ODMR before helium (determines if sensing is possible at all)
2. T_2^* at 10 mK via Ramsey
3. ^{4}He exposure response (non-magnetic reference)
4. ^{3}He matched exposure response
5. Isotope contrast $\Delta\Gamma = \Gamma_{^3\mathrm{He}} - \Gamma_{^4\mathrm{He}}$
6. Pressure/exposure-time sweep for τ_c bound
7. Sample temperature stability and residual gas baseline

Pass/fail logic: If $C_{\mathrm{contr}} = 0$: sensing fails; stop. If ^{3}He and ^{4}He both degrade equally: non-magnetic surface effect dominates. If ^{3}He uniquely increases decoherence above noise floor: candidate nuclear-spin signal. If $\tau_c < 292\,\mu$s from any extraction: detection forecast fails.

This experiment determines whether the sensing premise survives. Do not claim LCVD breakthrough before this measurement is performed.

19 Limitations

1. No physical measurement has been performed in this system.
2. All parameters are [**ASSUMED**], UNKNOWN, or from [**MANUFACTURER_SPEC**]/[**LITERAT**
3. Mode separation is specified; interlocks are not installed or tested.
4. τ_c on F-diamond at 10 mK is completely unknown.
5. C_{contr} at 10 mK is completely unknown.
6. G_{eff} is model-based; Ag sinter not fabricated.

7. Bibliography contains 23 entries but full-depth citation coverage of every numerical parameter requires further expansion.
8. LCVD at 10 mK is not demonstrated.
9. Kevlar/G-10CR support architecture is [**DESIGN_SPECIFIED**] and not fabricated.
10. QCM sub-Hz stability under cryogenic conditions is unverified.

20 Conclusion

This work does not demonstrate a working millikelvin LCVD/NV platform. It defines a proposed, mode-separated, same-chamber validation framework and the experimental program required to test it.

The current system is **blocked** because required hardware is not installed and required measurements are not verified.

Detection feasibility is governed by two co-equal scientific unknowns: $\tau_c \geq 292\,\mu$s (canonical v3.3) and $C_{\text{contr}} > 0$ at 10 mK. Neither is measured. Either alone is sufficient to prevent detection.

Engineering feasibility remains conditional on measured G_{eff}, residual gas suppression, η_{abs}, P_{mw}, S_{vib}, QCM stability, d_{NV}, n_s, surface termination, and interlock behavior.

Until those measurements are performed, this framework remains an **unvalidated conditional pre-digital-twin architecture** — not a demonstrated breakthrough.

Global status: Current: BLOCKED. Forecast (post-installation, Mode D): CONDITIONAL (33.8% hypothetical MC). Validated: not available. Breakthrough claim: not made.

A Superseded τ_c Thresholds

v3.0 threshold: $\tau_c \geq 27.7\,\mu$**s** [**SUPERSEDED v3.0**] The 27.7 [**SUPERSEDED v3.0**] μs value was derived for DC Ramsey sensing without pulse dephasing. It is superseded by the v3.3 combined threshold.

Canonical v3.3 threshold: $\tau_c \geq 292\,\mu$**s** Derived from: SNR$\geq$ 5 plus $\varepsilon_{\text{thermo}} < 1\%$, including pulse dephasing correction ($\tau_{\pi/2}/T_2^* = 1.13$; degraded regime). This is the value used in all gates, the final verdict, and the Monte Carlo.

v3.1 DC Ramsey threshold: $\tau_c \geq 27.7\,\mu$**s** [**SUPERSEDED v3.0**] Historical derivation retained for traceability. Not current gate logic.

Superseded AC model: $\tau_c \approx 65$ **ns** [**SUPERSEDED v3.0**] Used in earlier simulation versions. Not current.

B Rejected Vespel Baseline Calculations

Direct Vespel SP-22 supports are rejected from the committed architecture. For reference: three $\phi 3\,\mathrm{mm}$, $L = 50\,\mathrm{mm}$ rods would give $P_{\mathrm{support}} \approx 1.2\,\mathrm{nW}$ (no thermal interception), creating an unacceptable direct 4 K-to-10 mK conductive path. This is not the committed design.

C Hardware Registry

See `hardware_registry.json`. All items: [**DESIGN_SPECIFIED**] or [**NOT_INSTALLED**] unless marked otherwise. No item is VERIFIED.

D Validation Matrix

See `validation_matrix.csv` (157 rows; all `can_PASS_now = NO`).

E Source and Citation Audit

See `representative_source_audit.csv` and `source_audit_status.txt`. The included audit is REPRESENTATIVE_ONLY and not submission-complete. A full submission-grade numerical source audit remains incomplete.

F Simulation Output Schema

Run `python3 qta_full_sim.py` to regenerate all CSV/JSON outputs.

G Slop Removal Log

See `slop_removal_log.txt` for before/after record of removed overconfidence language.

Bibliography

References

[1] F. Pobell, *Matter and Methods at Low Temperatures*, 3rd ed. Springer (2007).

[2] E.T. Swartz and R.O. Pohl, Rev. Mod. Phys. **61**, 605 (1989).

[3] M. Barucci *et al.*, Cryogenics **45**, 295 (2005).

[4] C.L. Degen, F. Reinhard, and P. Cappellaro, Rev. Mod. Phys. **89**, 035002 (2017).

[5] L. Rondin *et al.*, Rep. Prog. Phys. **77**, 056503 (2014).

[6] J.R. Maze *et al.*, Nature **455**, 644 (2008).

[7] S. Kolkowitz *et al.*, Science **347**, 1129 (2015).

[8] M.E. Trusheim *et al.*, Phys. Rev. Lett. **124**, 020501 (2020).

[9] S. Sangtawesin *et al.*, Phys. Rev. X **9**, 031052 (2019).

[10] J.M. Taylor *et al.*, Nat. Phys. **4**, 810 (2008).

[11] B. Bernu *et al.*, Phys. Rev. Lett. **96**, 255503 (2006).

[12] D. Morhard *et al.*, Phys. Rev. B **53**, 14978 (1996).

[13] D.S. Greywall, Phys. Rev. B **47**, 309 (1993).

[14] G.A. Bird, *Molecular Gas Dynamics and the Direct Simulation of Gas Flows*, Oxford (1994).

[15] A. Roth, *Vacuum Technology*, 3rd ed., Elsevier (1990).

[16] J.F. O'Hanlon, *A User's Guide to Vacuum Technology*, 3rd ed., Wiley (2003).

[17] K. Jousten (ed.), *Handbook of Vacuum Technology*, Wiley-VCH (2016).

[18] CERN, *Outgassing Handbook*, 2nd ed. (2020).

[19] A. Dal Palmiero *et al.*, Vacuum **73**, 121 (2004).

[20] D. Giorgi *et al.*, J. Vac. Sci. Technol. A **28**, 1332 (2010).

[21] G. Sauerbrey, Z. Phys. **155**, 206 (1959).

[22] A. Ozkan *et al.*, Appl. Surf. Sci. **259**, 1 (2012).

[23] M. Shinoda *et al.*, Appl. Phys. A **94**, 305 (2009).